

Solutions to Worksheet #1

August 3, 2026

1. **Solution.** We use the action in which $\pi(x_i) = x_{\pi(i)}$. Write the three monomials of f as

$$A = x_1 x_2^2 x_4^4, \quad B = x_2^2 x_3^4 x_4, \quad C = x_1^4 x_2^2 x_3.$$

The variable x_2 is the only variable that has exponent 2 in every monomial. Therefore, any permutation that fixes f must fix 2.

The cycle $(1\ 4\ 3)$ fixes 2 and sends

$$A \mapsto B, \quad B \mapsto C, \quad C \mapsto A.$$

Thus $(1\ 4\ 3)$ fixes the sum f . Its inverse $(1\ 3\ 4)$ also fixes f , as does the identity.

There are no other possibilities. Indeed, after fixing 2, the image of A must be one of A , B , or C . Because the exponents 1, 0, and 4 on x_1, x_3, x_4 are distinct, each choice determines the permutation completely, and the three choices give precisely the permutations above. Hence

$$\boxed{\{\pi \in S_4 : \pi(f) = f\} = \{\text{id}, (1\ 4\ 3), (1\ 3\ 4)\}.$$

2. **Solution.** Start with the monomial $x_1^7 x_3^2$ and take its alternating sum over S_3 :

$$f = \sum_{\pi \in S_3} \text{sgn}(\pi) \pi(x_1^7 x_3^2).$$

Since the exponents 7, 2, and 0 are distinct, the six monomials in this sum are distinct. Expanding gives

$$f = x_1^7 x_3^2 - x_2^7 x_3^2 - x_1^2 x_3^7 - x_1^7 x_2^2 + x_1^2 x_2^7 + x_2^2 x_3^7.$$

In particular, $x_1^7 x_3^2$ occurs with coefficient 1.

To verify the required property, let $\sigma \in S_3$. Reindexing the sum with $\tau = \sigma\pi$ gives

$$\begin{aligned} \sigma(f) &= \sum_{\pi \in S_3} \text{sgn}(\pi) (\sigma\pi)(x_1^7 x_3^2) \\ &= \text{sgn}(\sigma) \sum_{\tau \in S_3} \text{sgn}(\tau) \tau(x_1^7 x_3^2) = \text{sgn}(\sigma) f. \end{aligned}$$

Alternative solution. The same polynomial can be written as the determinant

$$f = \begin{vmatrix} x_1^7 & x_2^7 & x_3^7 \\ 1 & 1 & 1 \\ x_1^2 & x_2^2 & x_3^2 \end{vmatrix}.$$

Permuting the variables permutes the columns of the determinant, so the determinant is multiplied by the sign of the permutation. Expanding the determinant also shows that the coefficient of $x_1^7 x_3^2$ is 1.

3. **Solution.** A monomial symmetric polynomial is the sum of all distinct monomials obtained by permuting the indicated exponents among the available variables. Therefore,

$$m_{3,2}(x_1, x_2) = x_1^3 x_2^2 + x_1^2 x_2^3,$$

and

$$\begin{aligned} m_{3,2}(x_1, x_2, x_3) &= x_1^3 x_2^2 + x_1^3 x_3^2 + x_2^3 x_1^2 \\ &\quad + x_2^3 x_3^2 + x_3^3 x_1^2 + x_3^3 x_2^2. \end{aligned}$$

The partition $(3, 3, 1, 1)$ has four positive parts, so it cannot be placed in only three variables. Hence

$$m_{3,3,1,1}(x_1, x_2, x_3) = 0.$$

With four variables, choose the two variables that receive exponent 3; the other two receive exponent 1. Thus

$$\begin{aligned} m_{3,3,1,1}(x_1, x_2, x_3, x_4) &= x_1^3 x_2^3 x_3 x_4 + x_1^3 x_3^3 x_2 x_4 + x_1^3 x_4^3 x_2 x_3 \\ &\quad + x_2^3 x_3^3 x_1 x_4 + x_2^3 x_4^3 x_1 x_3 + x_3^3 x_4^3 x_1 x_2. \end{aligned}$$

4. **Solution.** The coefficient of x^n in the product is obtained by the Cauchy product:

$$\begin{aligned} f(x)g(x) &= \sum_{n=0}^{\infty} \left(\sum_{\substack{i+j=n \\ i,j \geq 0}} j \right) x^n \\ &= \sum_{n=0}^{\infty} \left(\sum_{j=0}^n j \right) x^n = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n. \end{aligned}$$

Equivalently, as formal power series,

$$f(x) = \frac{1}{1-x}, \quad g(x) = \frac{x}{(1-x)^2},$$

so

$$\boxed{f(x)g(x) = \frac{x}{(1-x)^3} = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n.}$$

5. **Solution.** Distributing the two sums gives

$$\begin{aligned} fg &= \left(\sum_{i=1}^{\infty} x_i x_{i+1} \right) \left(\sum_{j=1}^{\infty} x_j \right) \\ &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} x_i x_{i+1} x_j. \end{aligned}$$

Separating the cases $j = i$, $j = i + 1$, and $j \notin \{i, i + 1\}$ gives the more explicit form

$$fg = \sum_{i=1}^{\infty} (x_i^2 x_{i+1} + x_i x_{i+1}^2) + \sum_{i=1}^{\infty} \sum_{\substack{j \geq 1 \\ j \neq i, i+1}} x_i x_{i+1} x_j.$$

The last double sum should not be simplified by discarding repetitions. For example, $x_i x_{i+1} x_{i+2}$ occurs twice: once using the adjacent pair $(i, i + 1)$ and once using the adjacent pair $(i + 1, i + 2)$.

6. **Solution.** By definition,

$$m_{2,1} = \sum_{i \neq j} x_i^2 x_j = \sum_{i < j} (x_i^2 x_j + x_i x_j^2).$$

On the other hand,

$$\begin{aligned} m_2 m_1 &= \left(\sum_i x_i^2 \right) \left(\sum_j x_j \right) \\ &= \sum_i x_i^3 + \sum_{i \neq j} x_i^2 x_j = m_3 + m_{2,1}. \end{aligned}$$

Therefore,

$$m_2 m_1 \neq m_{2,1}; \quad m_2 m_1 = m_3 + m_{2,1}.$$

7. **Solution.** In a product $m_3 m_2 m_1$, choose one variable for the exponent 3, one for the exponent 2, and one for the exponent 1. We classify the result according to which choices are equal.

- If all three choices are the same, we obtain m_6 .
- If only the choices for exponents 3 and 2 are the same, we obtain $m_{5,1}$.
- If only the choices for exponents 3 and 1 are the same, we obtain $m_{4,2}$.
- If only the choices for exponents 2 and 1 are the same, we obtain $m_{3,3}$. Each monomial of this type occurs twice, because either of its two variables can come from the m_3 factor.

- If all three choices are distinct, we obtain $m_{3,2,1}$.

Consequently,

$$m_3 m_2 m_1 = m_6 + m_{5,1} + m_{4,2} + 2m_{3,3} + m_{3,2,1}.$$

8. **Solution.** Recall that

$$m_{1^r} = \sum_{\substack{A \subseteq \{1,2,\dots\} \\ |A|=r}} \prod_{i \in A} x_i.$$

Fix one monomial of type $(2, 1^{a+b})$. It uses $a+b+1$ distinct variables: one variable is squared and the other $a+b$ variables occur to the first power.

To obtain this monomial from $m_{1^{a+1}} m_{1^{b+1}}$, the squared variable must be selected in both factors. Of the remaining $a+b$ variables, choose a to join it in the first factor; the remaining b must be selected in the second factor. There are

$$\binom{a+b}{a} = \binom{a+b}{b}$$

ways to do this. By symmetry every monomial in $m_{2,1^{a+b}}$ has the same coefficient. Therefore the requested coefficient is

$$\binom{a+b}{a}.$$

9. **Solution.**

- (a) In forming the product, we choose either 1 or x_i from each factor. A term obtained by choosing r distinct variables is a monomial of type (1^r) . Hence

$$\prod_{i=1}^{\infty} (1 + x_i) = \sum_{r=0}^{\infty} m_{1^r} = 1 + m_1 + m_{1,1} + m_{1,1,1} + \cdots$$

- (b) The same argument applies, but choosing $-x_i$ from r factors contributes the sign $(-1)^r$. Therefore,

$$\prod_{i=1}^{\infty} (1 - x_i) = \sum_{r=0}^{\infty} (-1)^r m_{1^r} = 1 - m_1 + m_{1,1} - m_{1,1,1} + \cdots$$

10. **Solution.** For each variable, use the geometric series

$$1 + x_i + x_i^2 + \cdots = \frac{1}{1 - x_i}.$$

When the products are multiplied out, every monomial $x_1^{a_1} x_2^{a_2} \cdots$ of finite total degree occurs exactly once. Grouping monomials whose positive exponents form the same partition λ gives

$$\prod_{i=1}^{\infty} \frac{1}{1-x_i} = 1 + \sum_{n=1}^{\infty} \sum_{\lambda \vdash n} m_{\lambda}.$$

The initial 1 corresponds to choosing 1 from every factor, or equivalently to the empty partition. Since the worksheet specifies partitions of positive integers, the requested sum is therefore

$$\sum_{n=1}^{\infty} \sum_{\lambda \vdash n} m_{\lambda} = \prod_{i=1}^{\infty} \frac{1}{1-x_i} - 1.$$

If the empty partition is included and its monomial symmetric function is defined to be 1, then the answer is simply $\prod_{i=1}^{\infty} (1-x_i)^{-1}$.